



# Using a Leading-Edge Macro-Cylinder as a Passive Improvement Method to Enhance the Stall Conditions of NACA 0012: An Experimental Study

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**Abstract:** Boundary layer separation at high angles of attack often limits the aerodynamic performance of airfoils. Flow control strategies are generally classified into active and passive methods, with the latter offering simple and energy-free solutions. In this study, a macro-cylinder with diameter of 4 mm and chord length of 300 mm was installed on the upper surface of a National Advisory Committee of Aeronautics (NACA) 0012 airfoil at different chord wise positions ( $X = 1, 2, 3$ , and  $3.5$  cm from the leading edge). NACA 0012 airfoil which has dimensions 150 mm chord and 300 mm span (symmetrical) Experiments were conducted in a subsonic wind tunnel at a free-stream velocity of 30 m/s and angles of attack ranging from  $0^\circ$  to  $16^\circ$  step 2. The results prove that Stall behavior was considerably changed by installing a state-of-the-art macro-cylinder. By energizing the boundary layer and postponing flow separation, the cylinder functioned as a passive vortex-like generator. The best overall configuration was obtained at  $X = 3.5$  cm. The maximum lift force reached 5.45 N at  $14^\circ$ , while the maximum lift coefficient ( $C_L$ ) reached 0.8378 at  $12^\circ$ . At  $16^\circ$ , the same configuration maintained a lift force of 5.38 N and  $C_L$  of 0.6715, indicating improved post-stall aerodynamic behavior compared with the baseline airfoil. This improvement is attributed to the macro-cylinder's ability to energize the boundary layer and suppress early separation.

**Keywords:** National Advisory Committee of Aeronautics 0012; Lift force; Drag force; Aerodynamic efficiency; Angle of attack

## 1 Introduction

Flow separation is an important aerodynamic phenomenon that causes low aerodynamics performance and airfoil stability at high angle of attacks (AOAs) of wings and airfoils. The poor lift at low angles of attack is due to the flow remaining close to the surface, hence the poor lift for symmetrical airfoils like the National Advisory Committee of Aeronautics (NACA) 0012. The pressure difference between the bottom and top of the airfoil becomes larger as the AOA gets closer to the critical AOA. This causes a sudden decrease in lift and an increase in drag caused by separation of the boundary layer [1, 2]. This is a key challenge for making small wind turbines, drones, and aircraft more efficient and safer at low Reynolds number [3, 4]. There are two types of flow control: efficient and inefficient. Without external power (as artificial jets, plasma engines, piezoelectric devices etc. do not work) efficient techniques are not possible for controlling [5]. Passive approach, however, relies on the use of vortex generators or geometric changes to control passive behavior of boundary layers, without any extra power input [6]. Passive techniques are especially appealing for small-scale applications because of their simplicity, cost-effectiveness and reliability [7].

## 2 Literature Review

Passive flow-control strategies have long been investigated to enhance airfoil aerodynamic performance, particularly to delay stall and suppress noise. Owls have garnered a lot of interest among bio-inspired strategies because of their capacity for quiet and effective flight in choppy environments [1, 2]. Rao and Liu [1] shown using computational models that serrations enhance aerodynamic stability in wind gusts. While Ikeda et al. [2] verified, using Particle Image Velocimetry (PIV) and force measurements whereby that serrations regulate vortex dynamics

for increased production of lift. Passive flow-control strategies have long been investigated to enhance airfoil aerodynamic performance, particularly to delay stall and suppress noise. Owls have garnered a lot of interest among bio-inspired strategies because of their capacity for quiet and effective flight in choppy environments. Together, their wings' downy surfaces, trailing-edge fringes, and leading-edge serrations lower aero acoustic emissions and change the behavior of the boundary layer [3, 4]. According to define the leading-edge serrations, which are critical to the preservation of aerodynamic strength in the presence of turbulence and gusts. Similar to this, Noda et al. [5] significantly reduced aerodynamic noise with thrust being maintained by serrated structures applied to drone propellers. Show that both analytical and numerical investigations ascertained that serrated leading edges can decrease tonal noise by disturbing which is consistent with sound emission characteristics of the coherent structures from an aeroacoustics perspective. Lyu and Azarpeyvand [6] developed for serrated airfoils predictions expressed little radiated turbulence noise streams. Geyer et al. [8] further established experimentally that the feather structures on the owl's comb suppress broadband noise well, whereas Mansour [9] reported that sinusoidal serrations mitigate turbulence–airfoil interaction noise. Morphological studies also suggested that serration geometry.

Passive separation-control devices like surface-mounted cylinders, vortex generators, and leading-edge modifications are in wide used. Also, have been extensively studied for improving airfoil performance near stall, as well as bio-inspired serrations. These devices can create controlled perturbations in the near-wall flow that can improve momentum transfer in the boundary layer and minimize the potential of premature separation. Passive devices are simpler, require no outside energy input, and are more appropriate for low-cost aerodynamic applications than are active flow-control techniques.

Wagner et al. [3] reviewed the difference in longitude and angle of serration and its spacing of owl species and their direct relationship with silent flight efficiency. The variation in serration shape has been shown to have aerodynamic significance by Mansour and Doos [10], who measured them in a number of species of owls. In addition, Nieto and Marin-Calvo [11] provided three-dimensional reconstructions of serrations and provided guidelines to be used in biomimetic applications in engineered airfoils.

The experimental aerodynamics was used to test these mechanisms on model planes. Serrations enhanced aerodynamic efficiency at low Reynolds numbers, demonstrated by owl-inspired wing sections by Winzen et al. [12] and Ito [13] with serrated small-scale wings. Klän et al. [14] extended this analysis with owl-based airfoil models and showed that, under adverse pressure gradients, there was improved flow attachment.

The history of using serration as noise and stall control measures can be traced back to the initial rotor and wing studies. The researches of Hersh et al. [15], Soderman [16], and Serrated leading edges are already being suggested by Collins [17] to reduce the low-speed rotor noise. These initial contributions pioneered the way to bioinspired methods of subsequent decades [18, 19]. Muhammad [20] conducted a study on the control of boundary layer separation using a cylinder placed at the leading edge of the NACA 0012 airfoil. The experimental work has been done using wind tunnel with air flow velocity 24 m/s, and with airfoil NACA 0012 at angles of attack ( $0^\circ$ ,  $5^\circ$ ,  $10^\circ$ ,  $14^\circ$ , and  $20^\circ$ ) respectively. The pressure distribution around airfoil NACA 0012 was measured at each above angles of attack to calculate lift and drag coefficients with and without circular cylinder at the leading edge of this airfoil. The results showed that, the angle of separation of the boundary layer of airfoil is  $14^\circ$ , but the angle will be change to  $20^\circ$  when the cylinder be at the leading edge of the airfoil.

Gnatowska and Gajewska [21] conducted experimental research using the PIV method on a NACA 0012 airfoil at a Reynolds number of 66,400. Initially, the airfoil was tested for three different angles of attack:  $13^\circ$ ,  $15^\circ$ ,  $17^\circ$ , and  $19^\circ$ . These tests revealed that angles of attack above  $15^\circ$  significantly increase boundary layer detachment. Then, in the second stage of the research, a different-shaped microcylinder with a characteristic dimension ( $d/c$ ) of 0.01 was added to the leading edge of the airfoil at a high AOA of  $17^\circ$ . Unlike traditional vortex generators placed at the rear of the airfoil, this configuration aimed to reduce boundary layer detachment. The experiment demonstrated that the microcylinder effectively reduced boundary layer detachment at AOA  $17^\circ$ .

The aerodynamic feedback of the NACA 0012 airfoil with the macro-cylinder however, is highly dependent upon the position of the cylinder in relation to the leading edge of the airfoil. A cylinder too near the leading edge might cause too much disturbance, and a cylinder further downstream might have a more beneficial interaction with the forming boundary layer. It is thus necessary to check multiple chordwise positions to determine which position(s) are most beneficial for lift increasing and stall delaying characteristics.

## 2.1 Research Gap

Although many passive flow-control techniques have been investigated for airfoil performance improvement. Most previous studies focused on fixed passive devices, serrated leading edges, or single-location surface modifications. Limited experimental attention has been given to the effect of macro-cylinder chordwise position on the aerodynamic behavior of NACA 0012 airfoils. In particular, the relationship between macro-cylinder location, lift-force development, lift-coefficient variation, and stall-delay behavior remains insufficiently clarified. Therefore, this study addresses this gap by experimentally testing several macro-cylinder positions, namely  $X = 1, 2, 3$ , and  $3.5$

cm from the leading edge, to identify the optimum location for improving lift performance and delaying stall.

## 2.2 Objective and Contribution

The main contribution of this work is the experimental evaluation of a leading-edge macro-cylinder as a passive flow-control device for improving the aerodynamic performance of a NACA 0012 airfoil. Unlike previous studies that focused mainly on fixed passive modifications or bio-inspired serrations. The present work compares several cylinder positions and identifies the most effective location based on lift force, lift coefficient, and stall behavior.

## 3 Experimental Setup

### 3.1 Subsonic Wind Tunnel

In order to study the interaction between an object and moving air, TecQuipment (TQ) subsonic wind tunnels with a  $300 \times 300 \times 600$  mm test section, single point balance, and 30 m/s maximum air flow was used. Figure 1 shows the TQ subsonic wind tunnel in the aerodynamic lab in the college of engineering of the Thi-Qar University. Eq. (1) was used to calculate the air velocity [1].

$$v = \sqrt{\frac{2 \times 9.81 \times \Delta h}{\rho a}} \quad (1)$$



Figure 1. Subsonic wind tunnel [1]

### 3.2 Test Model

NACA 0012 symmetrical cross-section developed by T-Q Equipment Company from England which has dimensions 150 mm chord and 300 mm span (symmetrical) was selected as the study's Airfoil. For plotting the profile of NACA 0012, data imported from an online airfoil tool was used. Figures 2, 3, and 4 schematic profiles of NACA 0012 airfoil without and with adding macro-cylinder modifications. Also, the improving process using in this study represented by fixing a macro-cylinder. The dimensions of macro-cylinder were ( $\Phi = 4$  mm,  $L = 300$  mm) at different chordwise positions ( $X = 1, 2, 3$ , and  $3.5$  cm) from leading edge of airfoil along the chord. The test section was experimented at different AOA and 30 m/s air speed. Table 1 shows the experimental parameters that were used in this study.

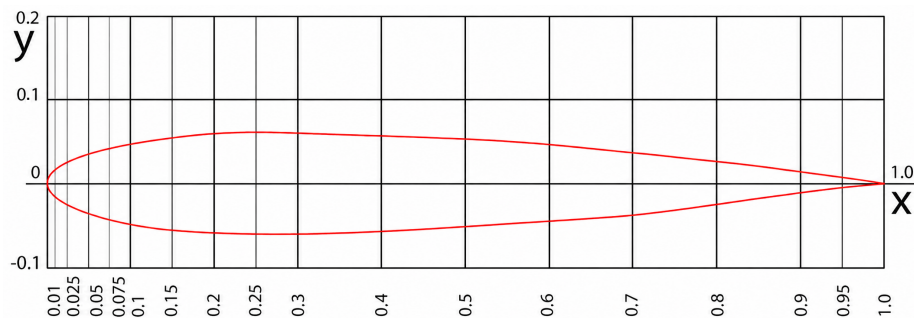
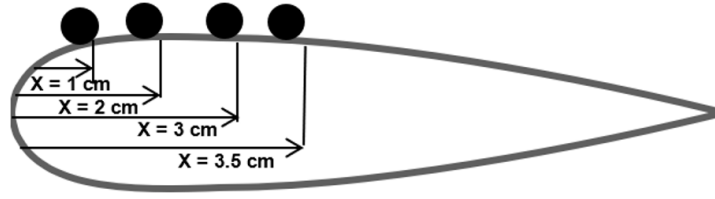
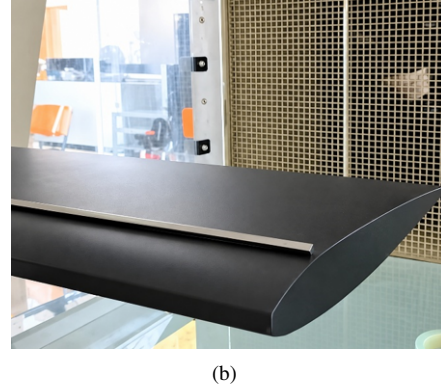
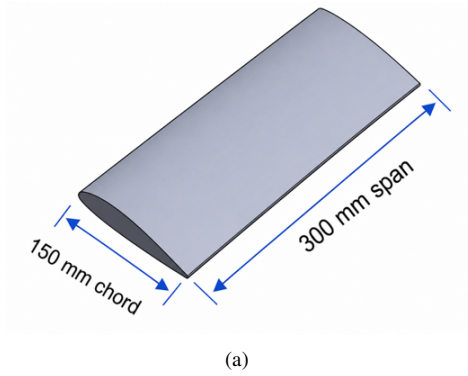


Figure 2. National Advisory Committee of Aeronautics (NACA) 0012 profile



**Figure 3.** National Advisory Committee of Aeronautics (NACA) 0012 macro-cylinder position with different distance on the surface airfoil



**Figure 4.** (a) National Advisory Committee of Aeronautics (NACA) 0012 airfoil; (b) macro-cylinder position on the surface airfoil in the laboratory

**Table 1.** Experimental parameters

| Description           | Symbol    | Amount                 | Units             |
|-----------------------|-----------|------------------------|-------------------|
| Angle attack          | $\alpha$  | (0–16°) step 2         | –                 |
| Free stream velocity  | $U_s$     | 30                     | m/s               |
| Air density           | $\rho$    | 1.2754                 | Kg/m <sup>3</sup> |
| Air dynamic viscosity | $\mu$     | $1.872 \times 10^{-5}$ | Kg/m·s            |
| Air temperature       | $T_a$     | 30                     | °C                |
| Atmosphere pressure   | $P_{atm}$ | 101.325                | kN/m <sup>2</sup> |

### 3.3 Theoretical Calculations

Several equations from text have been adopted to calculate the wing area, lift coefficient ( $C_L$ ), drag coefficient ( $C_D$ ), and lift/drag ratio ( $C_L/C_D$ ) in this study [20].

$$\text{Wing Area} = \text{chord} \times \text{span} \quad (2)$$

$$C_L = \frac{FL}{\frac{1}{2}\rho v^2 S} \quad (3)$$

$$C_D = \frac{FD}{\frac{1}{2}\rho v^2 S} \quad (4)$$

$$\text{Lift/Drag} = C_L/C_D \quad (5)$$

### 3.4 Boundary Conditions

All the experimental calculations were done at the following boundary conditions:

- The air velocity at input of the test section ( $v_\infty = 30$  m/s);
- Density of the air ( $\rho = 1.2754$  kg/m<sup>3</sup>);

- Air dynamic viscosity ( $\mu = 1.81 \times 10^{-5}$  kg/m·s);
  - Constant air temperature  $T_a = 298$  K);
  - The heat capacity ratio of the air ( $K = 1.4$ ).
- Then Reynolds number:

$$Re_c = \frac{\rho u_\infty c}{\mu} \quad (6)$$

### 3.5 Measurement Uncertainty and Repeatability

The experimental measurements were conducted under the same wind-tunnel operating conditions to ensure consistency. The free-stream velocity was kept constant at 30 m/s, and the airfoil model was tested over angles of attack from  $0^\circ$  to  $16^\circ$  with a  $2^\circ$  increment. The force readings were obtained using the wind-tunnel balance system. To improve the reliability of the reported data, the measurements were checked for repeatability under identical operating conditions, and the average readings were used in the analysis. The possible sources of uncertainty include balance resolution, angle-of-attack adjustment, air-speed fluctuation, and reading variation of lift and drag forces.

## 4 Results and Discussions

### 4.1 Experimental Results for National Advisory Committee of Aeronautics 0012 Without Macro-Cylinder Modification

The values of lift force ( $F_L$ ), drag force ( $F_D$ ),  $C_L$ ,  $C_d$ , and  $C_L/C_d$  at area of airfoil  $0.045 \text{ m}^2$  and density of the air ( $\rho = 1.2754 \text{ kg/m}^3$ ) are listed in Table 2.

**Table 2.** Experimental results without improving

| Angle of Attack (AOA) | $F_L$ (N) | $F_D$ (N) | $C_L$ | $C_D$ | $C_L/C_D$ |
|-----------------------|-----------|-----------|-------|-------|-----------|
| $0^\circ$             | 0.00      | 0.00      | 0.00  | 0.00  | 0.00      |
| $2^\circ$             | 0.56      | 0.36      | 0.08  | 0.025 | 3.20      |
| $4^\circ$             | 0.98      | 0.51      | 0.21  | 0.03  | 3.67      |
| $6^\circ$             | 1.75      | 0.61      | 0.35  | 0.04  | 8.75      |
| $8^\circ$             | 2.03      | 0.68      | 0.53  | 0.052 | 10.19     |
| $10^\circ$            | 2.66      | 0.74      | 0.61  | 0.055 | 11.09     |
| $12^\circ$            | 2.88      | 1.26      | 0.63  | 0.21  | 3.15      |
| $14^\circ$            | 2.70      | 1.52      | 0.53  | 0.23  | 2.30      |
| $16^\circ$            | 2.45      | 2.01      | 0.44  | 0.26  | 1.69      |

Notes:  $F_L$  = lift force;  $F_D$  = drag force;  $C_L$  = lift coefficient;  $C_D$  = drag coefficient;  $C_L/C_D$  = lift/drag ratio.

Variation of lift force with AOA is shown in Figure 5, from the figure it can be noted that,  $F_L$  increased when increasing the values of AOA and the maximum value was 2.88 N at  $\text{AOA} = 12^\circ$  because increasing airspeed and angle of attack dramatically increases lift forces. In addition, significantly decreasing and stall condition appeared in  $F_L$  values while  $\text{AOA} = 12^\circ$ . Figure 6 shows the influence AOA on the drag force. For this figure, the effect angles of attack on the drag forces small compare with lift force. This is because of the lack of attack area to air stream. At high range of angles of attack, there is significantly increased in the FD, as a result, to increase in the projected area that is front the stream.

Figure 7 illustrates the influence of AOA on  $C_L$  for the NACA 0012 airfoil. The  $C_L$  values increase gradually as the AOA increases, until it approaches its ultimate value at  $12^\circ$ . However, at an AOA of  $14^\circ$ , the  $C_L$  values decrease due to the occurrence of a stall condition. Due to the rapid development of distinct wake regions, the lift coefficient increases steadily with the AOA, particularly after the stall.

$C_D$  variation with AOA for NACA 0012 is exposed in the Figure 8. It can be gotten that, for this figure, the values of drag coefficient increased with increasing in values of AOA, due to the rise of the airspeed.

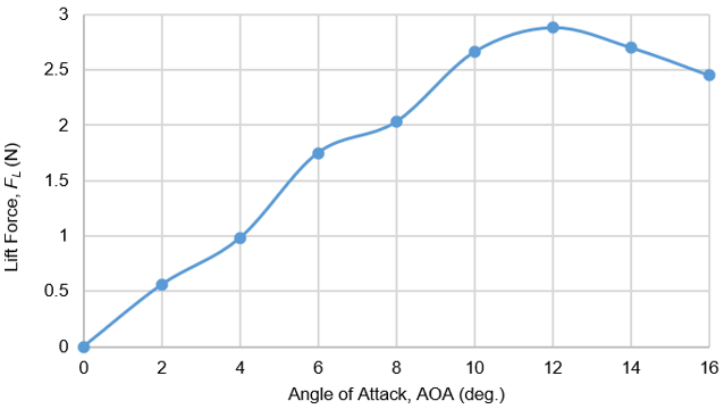
Figure 9 displays the relationship between  $C_L/C_D$  and attack angle (AOA). The increase in the lift and drag coefficient begins gradually until too that the maximum value of  $C_L/C_D = 11.09$  at  $\text{AOA} = 10^\circ$  and the state of begun decreasing at  $\text{AOA} = 12^\circ$ .

### 4.2 Experimental Results with Macro-Cylinder Modification for National Advisory Committee of Aeronautics 0012

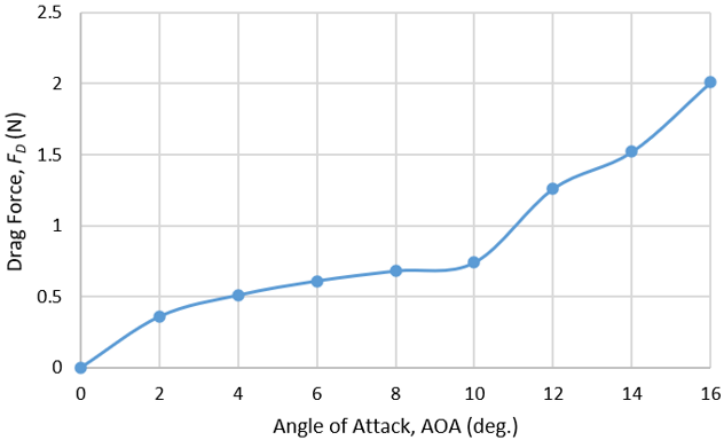
The values of  $F_L$  and  $C_L$  are listed in the Table 3 at distance ( $X = 1, 2, 3$ , and  $3.5 \text{ cm}$ ). A leading-edge macro-cylinder ( $L = 300 \text{ mm}$ ,  $\Phi = 4 \text{ mm}$ ) and density of air is ( $\rho = 1.2754 \text{ kg/m}^3$ ) fixed for all experiments.

Figure 10 shows the variation lift force with AOA for the NACA 0012 airfoil with adding macro- cylinder at different distance chordwise positions ( $X = 1, 2, 3$ , and  $3.5 \text{ cm}$ ). The lift force increased with the AOA up to the

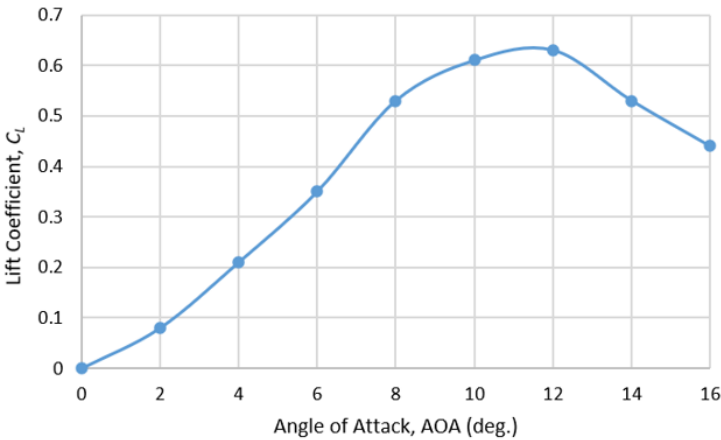
near-stall region. Among the tested positions,  $X = 3.5$  cm produced the best overall lift performance. The maximum lift force reached 5.45 N at  $14^\circ$ , while the lift force remained relatively high at 5.38 N at  $16^\circ$ . This indicates that the macro-cylinder at  $X = 3.5$  cm improved the post-stall lift behavior compared with the baseline airfoil.



**Figure 5.** Variation of lift force with Angle of Attack (AOA) for National Advisory Committee of Aeronautics (NACA) 0012 airfoil

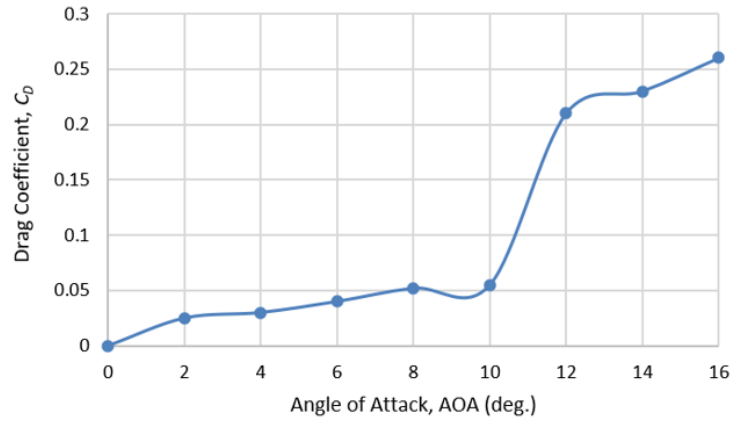


**Figure 6.** Variation of drag force with Angle of attack (AOA) for National Advisory Committee of Aeronautics (NACA) 0012 airfoil

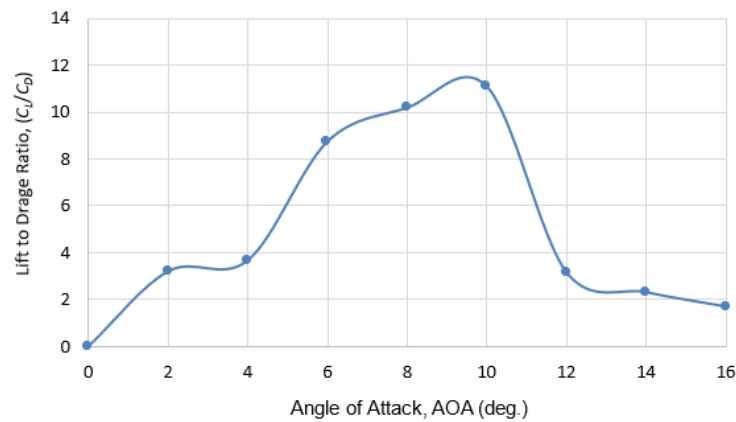


**Figure 7.** Variation of lift coefficient with Angle of Attack (AOA) for National Advisory Committee of Aeronautics (NACA) 0012 airfoil





**Figure 8.** Variation of drag coefficient with Angle of Attack (AOA) For National Advisory Committee of Aeronautics (NACA) 0012 airfoil



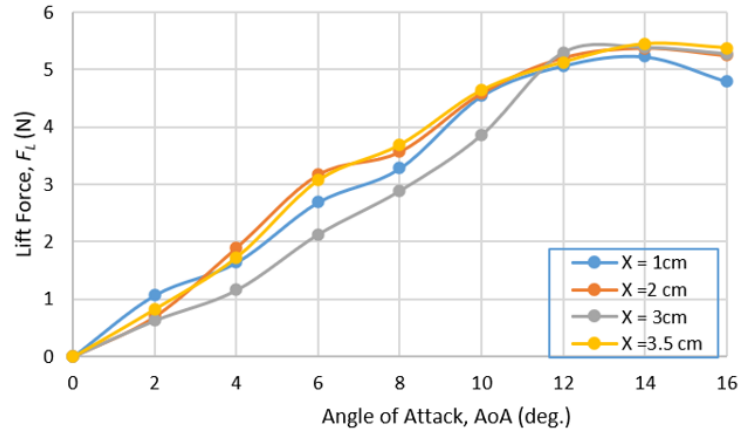
**Figure 9.** Variation of lift/drag ratio with Angle of Attack (AOA) for National Advisory Committee of Aeronautics (NACA) 0012 airfoil

**Table 3.** Experimental results with improving

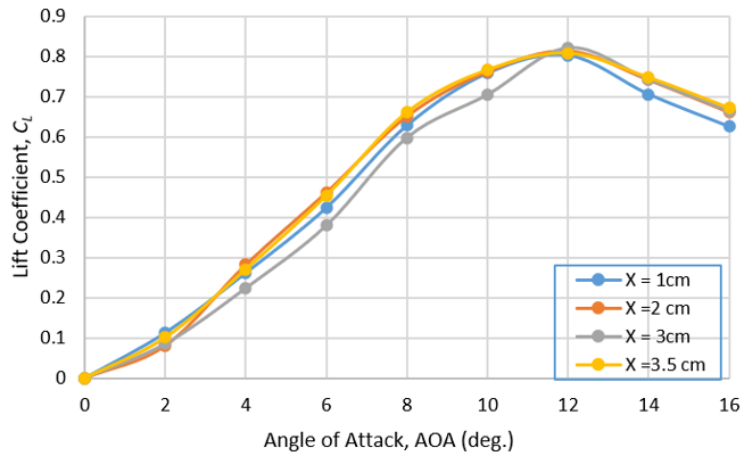
| Angle of Attack (AOA) | X = 1 cm  |        | X = 2 cm  |        | X = 3 cm  |        | X = 3.5 cm |        |
|-----------------------|-----------|--------|-----------|--------|-----------|--------|------------|--------|
|                       | $F_L$ (N) | $C_L$  | $F_L$ (N) | $C_L$  | $F_L$ (N) | $C_L$  | $F_L$ (N)  | $C_L$  |
| 0°                    | 0.00      | 0.0000 | 0.00      | 0.0000 | 0.00      | 0.0000 | 0.00       | 0.0000 |
| 2°                    | 1.07      | 0.1130 | 0.70      | 0.0911 | 0.63      | 0.0855 | 0.83       | 0.1013 |
| 4°                    | 1.64      | 0.2621 | 1.90      | 0.2827 | 1.16      | 0.2242 | 1.73       | 0.2712 |
| 6°                    | 2.69      | 0.4243 | 3.17      | 0.4622 | 2.13      | 0.3800 | 3.07       | 0.4543 |
| 8°                    | 3.28      | 0.6288 | 3.57      | 0.6517 | 2.89      | 0.5981 | 3.69       | 0.6612 |
| 10°                   | 4.54      | 0.7586 | 4.59      | 0.7625 | 3.86      | 0.7048 | 4.64       | 0.7665 |
| 12°                   | 5.07      | 0.8030 | 5.21      | 0.8141 | 5.30      | 0.8212 | 5.13       | 0.8378 |
| 14°                   | 5.23      | 0.7062 | 5.38      | 0.7418 | 5.39      | 0.7426 | 5.45       | 0.7473 |
| 16°                   | 4.80      | 0.6257 | 5.25      | 0.6613 | 5.28      | 0.6636 | 5.38       | 0.6715 |

Notes:  $F_L$  = lift force;  $C_L$  = lift coefficient.

The variation lift coefficient with AOA of NACA 0012 with the addition of macro-cylinder at various positions ( $X = 1, 2, 3$ , and  $3.5$  cm) on the surface air foil is shown in Figure 11. For this figure, it can illustrated that the lift coefficient increased with attack angle increasing. The maximum  $C_L$  was obtained at  $X = 3.5$  cm configuration, which was 0.8378 at  $12^\circ$ . Then the lift coefficient was reduced to 0.7473 at  $14^\circ$  and 0.6715 at  $16^\circ$  but the lift force remained high. Compared to the positions closer to the leading edge ( $X = 1$  cm or  $2$  cm), where disturbances may occur early in the flow causing local flow separation, the distance  $X = 3.5$  cm configuration is found to be a compromise between disturbance and reattachment. The above behaviour suggests the  $X = 3.5$  cm configuration led to better aerodynamic response near and outside the stall regime.



**Figure 10.** Variation of lift force with Angle of Attack (AOA) for National Advisory Committee of Aeronautics (NACA) 0012 airfoil at different distance from the surface



**Figure 11.** Variation of lift coefficient with Angle of Attack (AOA) for National Advisory Committee of Aeronautics (NACA) 0012 aerofoil at a different distance from the surface

### 4.3 Lift/Drag Ratio Trends after Stall

The results indicate a definite decrease in lift/drag ratio following the stall angle. This is behaviour consistent with conventional aerodynamic theory, whereby the boundary-layer separation leads to a reduction in lift generation and an increase in pressure drag. As the result of separated flow and the formation of a wake, the effective lifting capacity of the airfoil decreases during the post-stall regime. Thus, a decrease in  $C_L/C_D$  suggests the flow has gone from attached to separated flow.

### 4.4 Optimal Performance Mechanism at X = 3.5 cm

The improved performance at X = 3.5 cm may be attributed to a more favorable interaction between the macro-cylinder and the developing boundary layer. The improved performance at X = 3.5 cm may be attributed to a more favorable interaction between the macro-cylinder and the developing boundary layer. At this position, the cylinder may introduce controlled disturbances that enhance momentum exchange near the upper surface of the airfoil. This effect may help delay flow separation and maintain higher lift at larger angles of attack.

## 5 Conclusions

Based on the experimental and aerodynamic analyses conducted in this study, the following principal conclusions can be drawn:

1. The baseline NACA 0012 airfoil reached a maximum lift force of 2.88 N and a maximum lift coefficient of 0.63 at an AOA of 12°, after which the lift decreased, indicating the onset of stall.
2. The maximum  $C_L/C_D$  of the baseline airfoil was 11.09 at AOA = 10° followed by a reduction at higher angles of attack due to increasing drag and flow separation.



3. Installing a macro-cylinder on the upper surface of the airfoil improved the lift response compared with the baseline configuration, particularly near the stall and post-stall regions.

4. The best overall configuration was obtained at  $X = 3.5$  cm. The maximum lift force reached 5.45 N at  $14^\circ$ , while the maximum lift coefficient reached 0.8378 at  $12^\circ$ . At  $16^\circ$ , the same configuration maintained a lift force of 5.38 N and a lift coefficient of 0.6715, indicating improved post-stall aerodynamic behavior compared with the baseline airfoil.

5. Compared with the baseline airfoil, the maximum lift force increased from 2.88 N to 5.45 N, corresponding to an improvement of approximately 89.2%. The maximum lift coefficient increased from 0.63 to 0.8378, corresponding to an improvement of approximately 33.0%.

6. The modified configuration maintained improved aerodynamic performance up to  $16^\circ$ , while the baseline airfoil showed clear stall behavior after  $12^\circ$ . This indicates a stall-delay effect of approximately  $4^\circ$ .

### Author Contributions

Conceptualization, M.M.M. and A.A.O.; methodology, M.M.M. and A.A.O.; validation, A.A.O.; formal analysis, H.L.T. and Z.A.O.; investigation, H.L.T. and Z.A.O.; resources, M.M.M.; data curation, H.L.T. and Z.A.O.; writing—original draft preparation, H.L.T. and Z.A.O.; writing—review and editing, M.M.M. and A.A.O.; visualization, H.L.T.; supervision, M.M.M.; project administration, M.M.M. All authors have read and agreed to the published version of the manuscript.

### Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

### Conflicts of Interest

The authors declare that they have no conflicts of interest.

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## Nomenclature

|      |                                            |
|------|--------------------------------------------|
| AOA  | Angle of attack                            |
| APC  | Airfoil's performance coefficient          |
| NACA | National Advisory Committee of Aeronautics |

## Greek symbols

|        |                                  |
|--------|----------------------------------|
| $\rho$ | Air density (Kg/m <sup>3</sup> ) |
|--------|----------------------------------|

## Subscripts

|           |                                                     |
|-----------|-----------------------------------------------------|
| $c$       | Chord length of an airfoil (m)                      |
| $S$       | Area of an airfoil (m <sup>2</sup> )                |
| $C_D$     | Drag coefficient                                    |
| $C_L$     | Lift coefficient                                    |
| $C_L/C_D$ | Lift/drag ratio (Airfoil's performance coefficient) |
| $F_L$     | Lift force (N)                                      |
| $F_D$     | Drag force (N)                                      |
| $Re$      | Reynolds number                                     |
| $t$       | Airfoil's thickness (m)                             |